PYTHON FOUNDATIONS - ASSIGNMENT 07

Break & Continue

for loop + range() control flow

18 progressive Python challenges: stop a loop with break, skip an iteration with continue, and combine both in realistic scenarios.

for | range() | break | continue | if / else

Name: _____

Date: _____

Batch: _____

BEGINNER TRACK

18 CONTROL-FLOW CHALLENGES

Quick reference before you begin

Control the loop intentionally - do not use break or continue randomly.

Ask: Do I want to stop the entire loop, or only skip the current iteration?

break

Stop the entire loop immediately

Use when the required result is found.

continue

Skip only this iteration

The loop continues with the next value.

range()

start, stop, step

The stop value is excluded.

1

break

1, 2, 3, STOP

Loop ends completely.

2

continue

1, 2, skip 3, 4, 5

Only one iteration is skipped.

Important: Use only concepts already covered - input, arithmetic, logical/comparison operators, membership, if/elif/else, for loop, range(), break and continue. No functions required.

Stop the loop with break

Find the required condition and exit immediately.

1 Bus Seat Search

FOR RANGE BREAK

A passenger is looking for one specific seat in a bus.

1. Loop through seat numbers 1 to 40 using range().
2. The requested seat number is 23. When seat 23 is reached, print Requested seat found and stop the loop.
3. After the loop, print Search finished.

2 First Shared Multiple

FOR RANGE BREAK %

Find the earliest number that satisfies two divisibility rules.

1. Check numbers from 1 to 100.
2. Find the first number divisible by both 6 and 8. Print the number and immediately stop.
3. Do not continue checking larger numbers after the first match.

3 Low Battery Alert

FOR RANGE BREAK STEP

Simulate a battery percentage falling over time.

1. Generate battery values from 100 down to 0 in steps of 5.
2. Print each battery percentage until the value reaches 20.
3. At 20, print Low Battery Warning and stop the loop.

Stop the loop with break

Find the required condition and exit immediately.

4 First Passing Score

FOR RANGE BREAK IF

A teacher wants to identify the first passing score in a generated range.

1. Loop through marks from 30 to 60.
2. Passing starts at 40. Print the first mark that satisfies the passing rule.
3. Use break immediately after the first passing score is found.

5 Shopping Budget Crossing

FOR SUM BREAK

A customer keeps adding items until a spending limit is crossed.

1. Generate item prices from 100 to 1000 in steps of 100.
2. Keep adding each price to a running total.
3. As soon as the total becomes greater than 2000, print the current item price and total, then stop.

6 Square Threshold Finder

FOR RANGE BREAK **

Find the first number whose square crosses a threshold.

1. Check numbers from 1 to 20.
2. Calculate the square of each number.
3. Print and stop at the first number whose square is greater than 150.

Skip with continue

Ignore only the current value, then keep looping.

7 Maintenance Floor Skip

FOR RANGE CONTINUE

An elevator inspection should ignore floors closed for maintenance.

1. Loop through floors 1 to 20.
2. Skip floors 4, 8 and 13 using continue.
3. Print Inspection on floor X for every other floor.

8 Odd Ticket Processing

FOR RANGE CONTINUE %

A service desk should process only odd-numbered tickets.

1. Loop through ticket numbers 1 to 30.
2. If a ticket number is even, skip it using continue.
3. Print Processing ticket X only for odd tickets.

9 Skip Multiples of Three

FOR RANGE CONTINUE COUNT

Filter out values that match an exclusion rule.

1. Loop through numbers 1 to 50.
2. Skip every number divisible by 3.
3. Count and print how many numbers were actually processed.

Skip with continue

Ignore only the current value, then keep looping.

10 Qualifying Score Filter

FOR RANGE CONTINUE

Only scores at or above the qualifying mark should be processed.

1. Generate scores from 0 to 100 in steps of 5.
2. Skip every score below 40 using continue.
3. Print Qualified score: X for all remaining values.

11 Reserved Employee IDs

FOR RANGE CONTINUE %

Some generated employee IDs are reserved and must not be assigned.

1. Loop through IDs from 1001 to 1020.
2. Treat every ID divisible by 5 as reserved and skip it.
3. Print Assigned ID: X for all non-reserved IDs.

12 Delivery Day Scheduler

FOR RANGE CONTINUE

A delivery service works for two weeks but does not deliver every seventh day.

1. Loop through day numbers 1 to 14.
2. Skip days 7 and 14 using a logical or membership-style condition.
3. Print Delivery scheduled - Day X for every other day.

break + continue

Decide whether to skip one iteration or stop everything.

13 Warehouse Box Scan

FOR BREAK CONTINUE %

Scan boxes in order while ignoring damaged box numbers.

1. Loop through box numbers 1 to 50.
2. Skip boxes divisible by 6 because they are marked damaged.
3. When box 37 is reached, print Priority box found and stop the scan completely.

14 Game Level Progression

FOR BREAK CONTINUE LOGIC

A game skips bonus-only levels and stops when a target level is reached.

1. Loop through levels 1 to 30.
2. Skip every level divisible by 4 using continue.
3. When level 23 is reached, print Target level reached and stop the loop.

15 Security Checkpoint Route

FOR BREAK CONTINUE

A guard moves through checkpoints with some temporarily closed.

1. Loop through checkpoints 1 to 25.
2. Skip checkpoints 5, 10 and 15 because they are closed.
3. At checkpoint 22, print Final security checkpoint reached and stop.

break + continue

Decide whether to skip one iteration or stop everything.

16 Odd Number Hunt

FOR BREAK CONTINUE %

Find a specific kind of number without processing unnecessary values.

1. Loop through numbers 1 to 100.
2. Skip all even numbers using continue.
3. Among the remaining values, stop at the first number greater than 30 that is divisible by 7 and print it.

17 Customer Queue Control

FOR BREAK CONTINUE COUNT

A service counter skips unavailable customers but closes early for an emergency.

1. Loop through customer numbers 1 to 40.
2. Skip every 5th customer and do not count them as served.
3. When customer 28 is reached, print Emergency closure, stop the loop, and print the number of customers actually served.

18 Temperature Sensor Monitor

FOR BREAK CONTINUE IF

Process only warning temperatures and stop at a critical reading.

1. Generate temperatures from 20 to 60 in steps of 2.
2. Skip every temperature below 40 using continue. For values from 40 to 48, print Warning: X C.
3. At the first temperature of 50 or more, print Critical temperature and stop the loop.